JOHN JOSEPH ZAGARIAH DANIEL

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ML Engineer and Researcher with a Master's in Applied AI, hands-on experience designing evaluation frameworks for LLM hallucination mitigation, building end-to-end genomic prediction pipelines, and benchmarking vision-language models. 3+ years of enterprise software engineering experience building automation and data intelligence platforms. Seeking Applied ML and AI engineering roles to build scalable, production-grade intelligent systems.

TECHNICAL SKILLS

- **ML & AI:** PyTorch, HuggingFace Transformers, LoRA/PEFT, RAG, Agentic RAG, Scikit-learn, Optuna, LangChain, LangGraph, Sentence-Transformers
- **LLM & Agents:** Groq API, Google Gemini API, HuggingFace Inference API, Tool Calling, Function Calling, ReAct/Reflexion Agents, Guardrails, Multi-Agent Orchestration, Prompt Engineering, Synthetic Data Generation
- **Languages:** Python, Java, SQL, Shell, JavaScript
- **Frameworks & Tools:** FastAPI, Pydantic, Jina AI, Alembic, Next.js, Git, Splunk
- **Data & Infrastructure:** PostgreSQL, Redis, ChromaDB, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3, RDS, CloudWatch), OpenTofu, Docker, Vercel, GitHub Actions (CI/CD), HuggingFace Inference Endpoints
- **Security & Quality:** JWT Auth, bcrypt, Sentry, Opik, Ruff, MyPy

EXPERIENCE

Agriculture and Agri-Food Canada (Co-Op) — Ottawa, Ontario, Canada
Machine Learning Research Associate – Bioinformatics — Jan 2025 - Dec 2025

- Built an end-to-end, config-driven genomic prediction pipeline as a modular Python package, replacing fragmented research workflows with a reproducible architecture and cutting manual setup time by ~50%.
- Integrated traditional statistical, machine learning, deep learning, and graph-based models behind a unified interface, with automated data validation, experiment tracking, and Optuna-based hyperparameter tuning to improve reproducibility and accelerate model selection.
- Led development of analytical visualizations including gene-density maps and genetic-distance analyses that enabled clearer statistical interpretation and directly informed downstream breeding decisions.
- Co-authored 2 research papers, translating complex model outputs and statistical findings into clear written analysis for technical and scientific stakeholders.

Accenture Solution Pvt. Ltd — Chennai, Tamil Nadu, India
Packaged App Development Analyst — Aug 2021 – Dec 2023

- Designed an Azure cost-intelligence data pipeline using Azure Monitor, Data Factory, and Synapse to ingest, transform, and securely land cross-subscription resource usage data into Blob Storage for leadership budget planning.
- Engineered an internal Operational Readiness Testing platform adopted by 3 application teams, replacing spreadsheet workflows with Python automation, REST integrations, and custom Splunk commands to enable one-click ticket assignment.
- Built interactive real-time dashboards in Splunk and Power BI to centralize visibility across cloud spend, VM utilization, database health, security posture, and API performance metrics.
- Deployed serverless Azure Functions (Python) for automated, scheduled health checks on URL availability and VM uptime, strengthening incident response across the infrastructure estate.
- Configured advanced data parsing, event-handling, and CIM-compliant ingestion pipelines (props/transforms/inputs) across distributed UNIX, Linux, and Windows enterprise logging sources.
- Managed a high-availability distributed Splunk environment, using the Splunk Monitoring Console to proactively optimize cluster capacity, automate reporting via Python/Shell scripting, and integrate alerts with MS Teams webhooks.

PROJECTS

MedCliniQ: Benchmarking & Mitigating Hallucinations in Medical QA LLMs — Jan 2026 – Apr 2026

- Benchmarked 3 open-weight LLMs (BioMistral, Gemma, Llama-3.1) across 18 configurations on 3,000 questions; LoRA/RAG+LoRA architectures outperformed Zero-Shot/Role/RAG/RAG-CoV prompting (F1 > 0.89)
- Designed a dual evaluation framework analyzing F1 scores for closed tasks alongside multi-level (word, sentence, paragraph) semantic similarity metrics for open-ended clinical explanations.
- Shipped the top Gemma-7B-IT + LoRA configuration as a full-stack chatbot web application utilizing FastAPI, Streamlit, Firebase Auth/Firestore, and HuggingFace Inference Endpoints.

EDUCATION

University of Ottawa — Ontario, Canada
Master of Computer Science, Applied Artificial Intelligence — Jan 2024 – Apr 2026

Panimalar Engineering College, Anna University — Chennai, Tamil Nadu, India
Bachelor of Engineering, Computer Science and Engineering — Jun 2017 – Apr 2021

PUBLICATIONS

- You, F. M., et al. (2026). MultiGS: A comprehensive and user-friendly genomic prediction platform... Crop Breeding, Genetics and Genomics, 8(1), e260004. https://doi.org/10.20900/cbgg20260004
- You, F. M., et al. (2026). Genomic selection for seed yield enhances flax breeding efficiency. Molecular Breeding, 46(6), 61. https://doi.org/10.1007/s11032-026-01684-3 *Eliminated 61–91% of low-performing lines; cut field costs by 48–78%.*